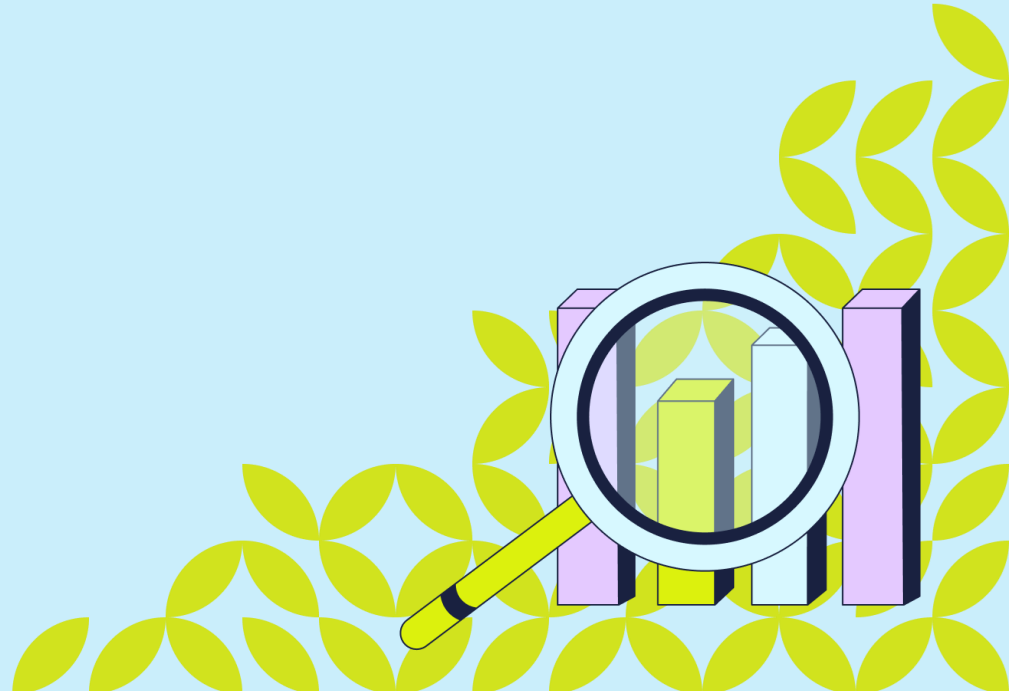


# Clockster

Work Documentation

Roy Padayao



# Project Overview

This project analyzes employee attendance data from HR department to identify employee discipline, violation trends, and department-level employee management.

The analysis required combining data from multiple sources and producing a shift-based discipline scoring model using SQL and Power BI.

Data source:

| Name                                                                                             | Date modified      | Type                  | Size   |
|--------------------------------------------------------------------------------------------------|--------------------|-----------------------|--------|
|  attendance     | 08/11/2022 4:23 pm | Microsoft Excel Co... | 695 KB |
|  leave_requests | 08/11/2022 4:23 pm | Microsoft Excel Co... | 5 KB   |
|  payroll        | 08/11/2022 4:23 pm | Microsoft Excel Co... | 34 KB  |
|  schedules      | 20/01/2025 7:16 pm | Microsoft Excel Co... | 627 KB |
|  users          | 08/11/2022 4:23 pm | Microsoft Excel Co... | 7 KB   |

## Raw data initial exploration.

- Inspection of column types, date formats, duplication, and missing values.

## Attendance

| user_id | first_name | last_name | location  | date       | time     | timezone | case | source |
|---------|------------|-----------|-----------|------------|----------|----------|------|--------|
| 74025   |            |           |           | 20/10/2021 | 20:08:45 | +08:00   | IN   | mobile |
| 74025   |            |           |           | 20/10/2021 | 20:09:11 | +08:00   | IN   | mobile |
| 74027   |            |           |           | 20/10/2021 | 22:29:56 | +08:00   | IN   | mobile |
| 74027   |            |           |           | 21/10/2021 | 0:15:33  | +08:00   | OUT  | mobile |
| 74025   |            |           | Nu Orange | 21/10/2021 | 13:22:20 | +08:00   | IN   | mobile |
| 74027   |            |           | Nu Orange | 21/10/2021 | 15:47:52 | +08:00   | IN   | mobile |
| 74135   |            |           | Nu Orange | 21/10/2021 | 18:09:37 | +08:00   | IN   | mobile |
| 74025   |            |           | Nu Orange | 22/10/2021 | 13:40:57 | +08:00   | IN   | mobile |
| 74138   |            |           |           | 22/10/2021 | 14:33:25 | +08:00   | IN   | mobile |
| 74027   |            |           | Nu Orange | 22/10/2021 | 16:15:48 | +08:00   | IN   | mobile |
| 74050   |            |           | Clinic    | 25/10/2021 | 13:51:35 | +08:00   | IN   | mobile |
| 74027   |            |           |           | 25/10/2021 | 13:58:21 | +08:00   | IN   | mobile |
| 74053   |            |           | Clinic    | 25/10/2021 | 14:20:04 | +08:00   | IN   | mobile |
| 74049   |            |           | Clinic    | 25/10/2021 | 14:20:16 | +08:00   | IN   | mobile |
| 74050   |            |           |           | 25/10/2021 | 14:20:21 | +08:00   | IN   | mobile |
| 74465   |            |           | Nu Orange | 25/10/2021 | 14:20:25 | +08:00   | IN   | mobile |
| 74054   |            |           | Nu Orange | 25/10/2021 | 14:20:29 | +08:00   | IN   | mobile |
| 74556   |            |           | Nu Orange | 25/10/2021 | 14:20:29 | +08:00   | IN   | mobile |
| 74053   |            |           | Clinic    | 25/10/2021 | 14:20:44 | +08:00   | IN   | mobile |

| user_id | first_name | last_name | type  | leave_type   | dates                                                                                                                                                     | time_start | time_end | timezone | status   | created_at       |
|---------|------------|-----------|-------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|----------|----------|------------------|
| 74463   |            |           | leave | day_off      | "2021-10-27"]                                                                                                                                             |            |          |          | accepted | 25/10/2021 12:38 |
|         |            |           | leave | day_off      | "2021-11-12"]                                                                                                                                             |            |          |          | accepted | 07/11/2021 15:20 |
| 74463   |            |           | leave | unpaid       | "2021-11-16"]                                                                                                                                             |            |          |          | accepted | 15/11/2021 13:45 |
| 75839   |            |           | leave | day_off      | "2021-11-24"]                                                                                                                                             |            |          |          | accepted | 16/11/2021 2:18  |
| 74042   |            |           | leave | day_off      | "2021-11-28"]                                                                                                                                             |            |          |          | accepted | 27/11/2021 7:56  |
| 74054   |            |           | leave | sick         | "2021-12-23"]                                                                                                                                             |            |          |          | accepted | 23/12/2021 4:54  |
| 74049   |            |           | leave | sick         | "2021-12-09", "2021-12-10", "2021-12-11", "2021-12-12", "2021-12-13", "2021-12-14"]                                                                       |            |          |          | accepted | 27/12/2021 13:50 |
| 74044   |            |           | leave | sick         | "2022-01-07", "2022-01-08"]                                                                                                                               |            |          |          | accepted | 07/01/2022 2:25  |
| 74042   |            |           | leave | annual       | "2022-03-11"]                                                                                                                                             |            |          |          | accepted | 01/03/2022 7:45  |
| 74461   |            |           | leave | compensatory | "2022-06-30"]                                                                                                                                             |            |          |          | accepted | 28/05/2022 7:98  |
| 83885   |            |           | leave | annual       | "2022-06-03"]                                                                                                                                             |            |          |          | accepted | 28/05/2022 7:49  |
| 88348   |            |           | leave | day_off      | "2022-06-07", "2022-06-08", "2022-06-09"]                                                                                                                 |            |          |          | accepted | 03/06/2022 10:92 |
| 83902   |            |           | leave | day_off      | "2022-06-07"]                                                                                                                                             |            |          |          | accepted | 03/06/2022 12:25 |
| 83902   |            |           | leave | day_off      | "2022-06-17"]                                                                                                                                             |            |          |          | accepted | 09/06/2022 4:39  |
| 88348   |            |           | leave | day_off      | "2022-06-14"]                                                                                                                                             |            |          |          | accepted | 13/06/2022 9:22  |
| 74042   |            |           | leave | compensatory | "2022-06-28", "2022-06-29"]                                                                                                                               |            |          |          | accepted | 15/06/2022 8:33  |
| 88357   |            |           | leave | sick         | "2022-06-23", "2022-06-24"]                                                                                                                               |            |          |          | accepted | 23/06/2022 13:04 |
| 74048   |            |           | leave | sick         | "2022-06-20", "2022-06-21", "2022-06-22", "2022-06-23", "2022-06-24", "2022-06-25", "2022-06-26", "2022-06-27", "2022-06-28", "2022-06-29", "2022-06-30"] |            |          |          | accepted | 27/06/2022 22:14 |
| 74042   |            |           | leave | compensatory | "2022-06-30"]                                                                                                                                             |            |          |          | accepted | 29/06/2022 18:27 |
| 88357   |            |           | leave | day_off      | "2022-07-11"]                                                                                                                                             |            |          |          | accepted | 11/07/2022 0:47  |
| 88357   |            |           | leave | compensatory | "2022-07-12", "2022-07-13"]                                                                                                                               |            |          |          | accepted | 12/07/2022 13:11 |
| 88357   |            |           | leave | unpaid       | "2022-07-14"]                                                                                                                                             |            |          |          | accepted | 14/07/2022 1:47  |
| 74461   |            |           | leave | compensatory | "2022-07-15", "2022-07-16", "2022-07-17", "2022-07-18", "2022-07-19", "2022-07-20"]                                                                       |            |          |          | accepted | 18/07/2022 8:28  |
| 88357   |            |           | leave | sick         | "2022-07-15", "2022-07-16", "2022-07-17", "2022-07-18", "2022-07-19", "2022-07-20"]                                                                       |            |          |          | rejected | 19/07/2022 11:20 |
| 74042   |            |           | leave | compensatory | "2022-08-01", "2022-08-02"]                                                                                                                               |            |          |          | accepted | 22/07/2022 8:44  |
| 83902   |            |           | leave | day_off      | "2022-08-29"]                                                                                                                                             |            |          |          | accepted | 26/07/2022 4:16  |
| 74461   |            |           | leave | compensatory | "2022-09-27", "2022-09-28", "2022-09-29", "2022-09-30"]                                                                                                   |            |          |          | accepted | 30/07/2022 3:35  |
| 75763   |            |           | leave | annual       | "2022-08-07"]                                                                                                                                             |            |          |          | accepted | 01/08/2022 9:04  |
| 74052   |            |           | leave | compensatory | "2022-08-06"]                                                                                                                                             |            |          |          | accepted | 01/08/2022 9:43  |
| 74049   |            |           | leave | compensatory | "2022-08-13"]                                                                                                                                             |            |          |          | accepted | 09/08/2022 23:54 |
| 88348   |            |           | leave | day_off      | "2022-08-10"]                                                                                                                                             |            |          |          | accepted | 09/08/2022 2:31  |
| 74042   |            |           | leave | compensatory | "2022-08-20"]                                                                                                                                             |            |          |          | accepted | 13/08/2022 1:51  |
| 120666  |            |           | leave | compensatory | "2022-08-21"]                                                                                                                                             |            |          |          | accepted | 16/08/2022 8:18  |
| 83885   |            |           | leave | compensatory | "2022-08-27"]                                                                                                                                             |            |          |          | accepted | 18/08/2022 8:57  |

## Schedules

[illegible]

# SQL

## Data cleaning using SQL

- Unnest and Splitting columns to rows

Query Query History

```
1 WITH leave_req1_w_row_num AS (  
2   SELECT DISTINCT *,  
3   ROW_NUMBER() OVER() AS row  
4 )  
5 )  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18
```

Data Output Messages Notifications

| user_id | first_name | last_name | type  | leave_type   | dates                                | time_start | time_end | timezone | status   | created_at |
|---------|------------|-----------|-------|--------------|--------------------------------------|------------|----------|----------|----------|------------|
| 74049   |            |           | leave | compensatory | [2022-09-13]                         | [null]     | [null]   | [null]   | accepted | 2022-09-08 |
| 88357   |            |           | leave | unpaid       | [2022-09-08, 2022-09-09, 2022-09-10] | [null]     | [null]   | [null]   | accepted | 2022-09-08 |
| 79765   |            |           | leave | annual       | [2022-09-20]                         | [null]     | [null]   | [null]   | accepted | 2022-09-10 |
| 79738   |            |           | leave | compensatory | [2022-09-28]                         | [null]     | [null]   | [null]   | accepted | 2022-09-10 |
| 74042   |            |           | leave | compensatory | [2022-09-28]                         | [null]     | [null]   | [null]   | accepted | 2022-09-10 |
| 74042   |            |           | leave | compensatory | [2022-10-11, 2022-10-12]             | [null]     | [null]   | [null]   | accepted | 2022-10-10 |
| 88348   |            |           | leave | annual       | [2022-10-10]                         | [null]     | [null]   | [null]   | accepted | 2022-10-10 |
| 74052   |            |           | leave | compensatory | [2022-10-09]                         | [null]     | [null]   | [null]   | accepted | 2022-10-10 |
| 74049   |            |           | leave | compensatory | [2022-10-14]                         | [null]     | [null]   | [null]   | accepted | 2022-10-10 |
| 83884   |            |           | leave | compensatory | [2022-10-22, 2022-10-23]             | [null]     | [null]   | [null]   | accepted | 2022-10-10 |
| 74049   |            |           | leave | compensatory | [2022-10-21]                         | [null]     | [null]   | [null]   | pending  | 2022-10-10 |

Total rows: 51 of 51 Query complete 00:00:00.062 Ln 2, Col 17

Query Query History

```
1 WITH leave_req1_w_row_num AS (  
2   SELECT DISTINCT *,  
3   ROW_NUMBER() OVER() AS row  
4   FROM leave_req1  
5 ),  
6 dates_table AS (  
7   SELECT row, UNNEST(string_to_array(dates, ',')) AS date  
8   FROM leave_req1_w_row_num  
9 )  
10 SELECT  
11   l.user_id,  
12   l.type,  
13   l.leave_type,  
14   l.status,  
15   l.created_at,  
16   to_date(REPLACE(REPLACE(dt.date, '[', ''), ']', ''), 'YYYY-MM-DD') AS date  
17 FROM leave_req1_w_row_num l  
18 LEFT JOIN dates_table dt ON dt.row = l.row;
```

Data Output Messages Notifications

|   | user_id | type  | leave_type   | status   | created_at          | date       |
|---|---------|-------|--------------|----------|---------------------|------------|
| 1 | 74044   | leave | sick         | accepted | 2022-01-07 02:25:11 | 2022-01-07 |
| 2 | 74044   | leave | sick         | accepted | 2022-01-07 02:25:11 | 2022-01-08 |
| 3 | 83885   | leave | compensatory | accepted | 2022-08-18 08:57:08 | 2022-08-27 |
| 4 | 83902   | leave | day_off      | accepted | 2022-07-26 04:16:11 | 2022-07-29 |
| 5 | 88357   | leave | sick         | rejected | 2022-07-19 11:20:32 | 2022-07-15 |

Total rows: 92 of 92 Query complete 00:00:00.060

# Normalization

- Collapse duplicate logs per day
- Joining tables
- Creating Shift-Level Flags

| Flag                   | Meaning                                      |
|------------------------|----------------------------------------------|
| counted_day_flag       | Scheduled day AND not accepted/pending leave |
| excused_flag           | Accepted/Pending leave                       |
| late_in_flag           | Actual in > 10 mins AND < 2 hours            |
| early_out_flag         | Actual out < -10 mins AND > -2 hours         |
| unexcused_absence_flag | No in/out AND not excused                    |
| incomplete_punch_flag  | Only one of IN/OUT present                   |

The screenshot shows a SQL IDE interface with a query editor on the left and a data output table on the right. The query is a complex SQL statement that joins several tables and calculates flags based on attendance data.

```
WITH daily AS (  
  SELECT  
    s.user_id,  
    s.date::date AS work_date,  
    s.sched_type,  
    s.time_start,  
    s.time_end,  
  
    lr.leave_type,  
    lr.status,  
  
    MIN(a.time) FILTER (WHERE a.casel = 'IN') AS in_time,  
    MAX(a.time) FILTER (WHERE a.casel = 'OUT') AS out_time  
  
  FROM clean_schedules s  
  LEFT JOIN attendance a  
    ON s.user_id = a.user_id  
    AND s.date = a.date  
  LEFT JOIN clean_leave_req lr  
    ON s.user_id = lr.user_id  
    AND s.date = lr.date  
  
  WHERE s.sched_type = 'work'  
  AND s.date BETWEEN '2021-10-20' AND '2022-10-19'  
  GROUP BY s.user_id, s.date, s.time_start, s.time_end, lr.leave_type, lr.status, s.sched_type  
)  
flags AS (  
  SELECT  
    user_id,  
    work_date,
```

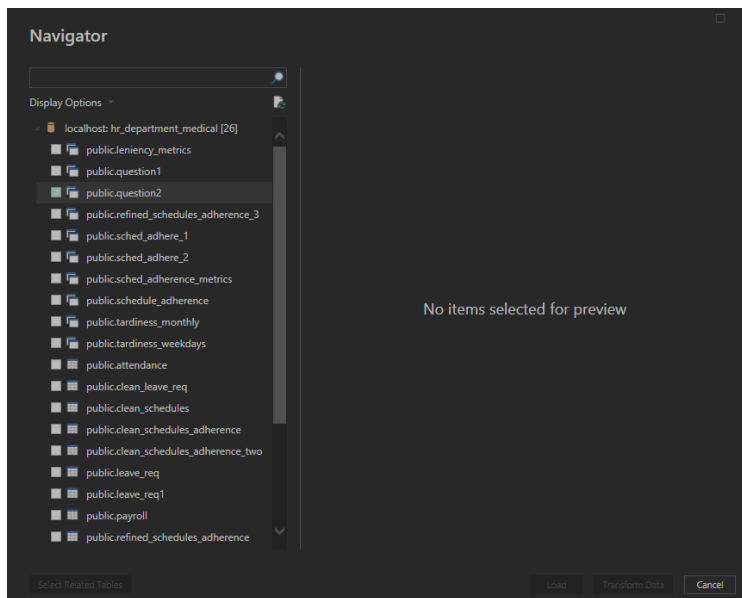
The data output table shows the results of the query, with columns for user\_id, work\_date, sched\_type, time\_start, time\_end, leave\_type, leave\_status, in\_time, out\_time, excused\_flag, and counted\_day\_flag. The table displays 5 rows of data, with the first row being a header row and the subsequent rows containing specific data points.

| ser_id | work_date  | sched_type | time_start | time_end | leave_type | leave_status | in_time  | out_time | excused_flag | counted_day_flag |
|--------|------------|------------|------------|----------|------------|--------------|----------|----------|--------------|------------------|
| 20666  | 2022-05-09 | work       | 15:00:00   | 22:00:00 | [null]     | none         | 14:52:55 | 22:46:38 | false        | true             |
| 20666  | 2022-05-10 | work       | 15:00:00   | 22:00:00 | [null]     | none         | 14:48:26 | 22:14:48 | false        | true             |
| 20666  | 2022-05-11 | work       | 08:00:00   | 15:00:00 | [null]     | none         | 07:56:48 | 15:01:54 | false        | true             |
| 20666  | 2022-05-12 | work       | 08:00:00   | 15:00:00 | [null]     | none         | 14:47:31 | 22:02:45 | false        | true             |
| 20666  | 2022-05-12 | work       | 15:00:00   | 22:00:00 | [null]     | none         | 14:47:31 | 22:02:45 | false        | true             |

Total rows: 13395 Query complete 00:00:00.585

# Data Modeling

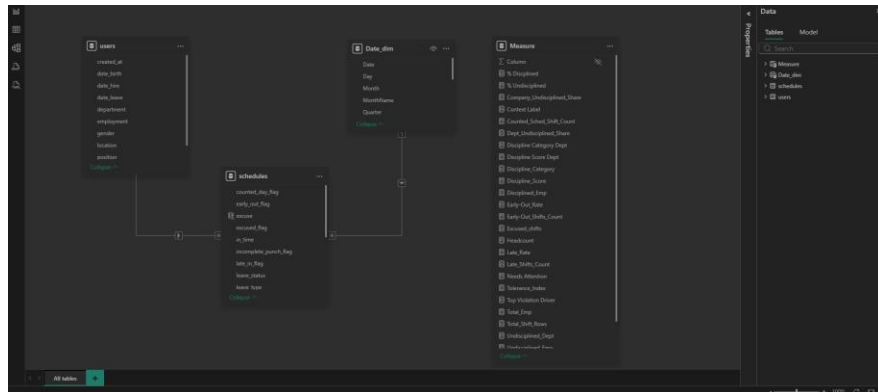
- Importing the transformed and shaped data tables from SQL to Power BI



No items selected for preview

The screenshot shows the Power BI Query Editor interface. The top ribbon includes tabs for Home, Transform, Add Column, View, Tools, and Help. The main area displays a SQL query: `= Value.FromQuery(PostgreSQL.Database("localhost"), "hr_department_medical"), select * from`. Below the query, a table of results is shown with columns: user\_id, work\_date, sched\_type, time\_start, time\_end, and leave\_type. The table contains 21 rows of data, showing various work dates and times. The bottom status bar indicates '18 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

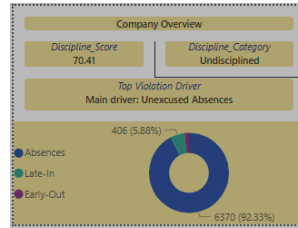
| user_id | work_date  | sched_type | time_start | time_end    | leave_type |
|---------|------------|------------|------------|-------------|------------|
| 1       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 2       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 3       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 4       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 5       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 6       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 7       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 8       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 9       | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 10      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 11      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 12      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 13      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 14      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 15      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 16      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 17      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 18      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 19      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 20      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |
| 21      | 12/06/2022 | work       | 8:00:00 am | 10:00:00 pm |            |



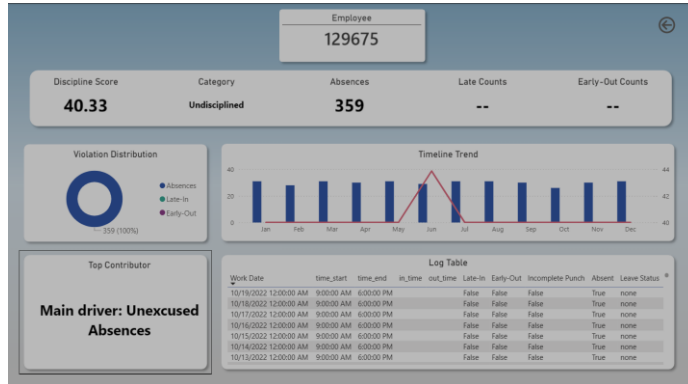
# Visualization

- Creating dashboard to visualize and analyze data.

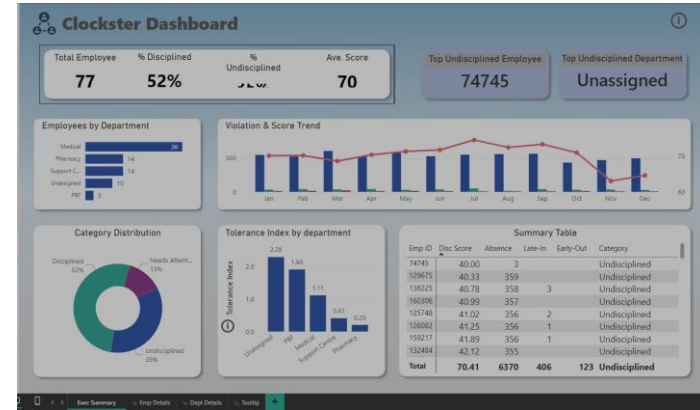
## Tooltip



## Employee Drillthrough page



## Executive Summary



## Department Drillthrough page

